

```
from google.colab import files
import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Upload image
uploaded = files.upload()
file_name = next(iter(uploaded))

# 2. Read image in color and convert to grayscale
img_color = cv2.imread(file_name)
gray = cv2.cvtColor(img_color, cv2.COLOR_BGR2GRAY)

# 3. Define Prewitt kernels
prewitt_x = np.array([
    [-1, 0, 1],
    [-1, 0, 1],
    [-1, 0, 1]
], dtype=np.float32)

prewitt_y = np.array([
    [-1, -1, -1],
    [ 0,  0,  0],
    [ 1,  1,  1]
], dtype=np.float32)

# 4. Define Sobel kernels
sobel_x = np.array([
    [-1, 0, 1],
    [-2, 0, 2],
    [-1, 0, 1]
], dtype=np.float32)

sobel_y = np.array([
    [-1, -2, -1],
    [ 0,  0,  0],
    [ 1,  2,  1]
], dtype=np.float32)

# 5. Apply Prewitt operator
prewitt_x_result = cv2.filter2D(gray, cv2.CV_32F, prewitt_x)
prewitt_y_result = cv2.filter2D(gray, cv2.CV_32F, prewitt_y)

prewitt = cv2.magnitude(
    prewitt_x_result,
    prewitt_y_result
)

# 6. Apply Sobel operator
sobel_x_result = cv2.filter2D(gray, cv2.CV_32F, sobel_x)
sobel_y_result = cv2.filter2D(gray, cv2.CV_32F, sobel_y)

sobel = cv2.magnitude(
    sobel_x_result,
    sobel_y_result
)

# 7. Convert results to 8-bit for display
prewitt = cv2.convertScaleAbs(prewitt)
sobel = cv2.convertScaleAbs(sobel)

# 8. Display results
plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
```

```
plt.imshow(cv2.cvtColor(img_color, cv2.COLOR_BGR2RGB))
```

```
plt.title("Original Color Image")
```

```
plt.axis("off")
```

```
plt.subplot(2, 2, 2)
```

```
plt.imshow(rewitt, cmap="gray")
```

```
plt.title("Prewitt Edge Detection")
```

```
plt.axis("off")
```

```
plt.subplot(2, 2, 3)
```

```
plt.imshow(sobel, cmap="gray")
```

```
plt.title("Sobel Edge Detection")
```

```
plt.axis("off")
```

```
plt.subplot(2, 2, 4)
```

```
difference = cv2.absdiff(rewitt, sobel)
```

```
plt.imshow(difference, cmap="gray")
```

```
plt.title("Difference: Sobel vs Prewitt")
```

```
plt.axis("off")
```

```
plt.tight_layout()
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```
plt.show()
```

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Original Color Image



Prewitt Edge Detection

